Study Guide — The Physicist's Toolkit

Physics 2026–27 · work in class Thursday, September 10 · finish for homework · due at the Skills Check, Monday, September 14

NAME _____ DATE _____

How to work this guide. The sections below are a ladder — each rung asks more of you than the last, and **Monday's Skills Check climbs the same ladder**. Sections 1–3 are class work for Thursday: work them with your table group, check answers as they come, and flag anything that fights back. Section 4 is homework — finish it before Monday and bring this guide to the check, because it is due at the door. Every quantitative problem runs the full protocol: labeled **K / D / E / A**, units traveling with the math, symbols rearranged before numbers go in. While you work, keep your draft **4×6 card** next to you — anything you have to look up twice belongs on it (see *How to Build a Cheat Sheet*).

Rung 1 — Warm-Up: Recall Only

SOLO · ~20 MIN · NO PROTOCOL NEEDED YET

If a rung-1 item costs you more than a moment, that's a flashcard for your card, not a mystery.

1 Back to base

UNITS & PREFIXES

Convert each measurement to the SI base unit. Follow the four steps from class: name the prefix, write its power of ten, multiply, **sanity-check the direction**.

a. A 3.5 km morning jog, in meters

b. A 280 ms blink of an eye, in seconds

c. A 68 µm dust grain, in meters (scientific notation is fine here)

d. Which is larger, 7×10^{-3} s or 9×10^{-4} s? Answer without writing either out in full — say how you knew.

WORK

2 Count & convert

SIG FIGS · NOTATION

a. Count the significant figures: $0.00380\ \text{kg} \cdot 4017\ \text{m} \cdot 9.90 \times 10^{6}\ \text{s} \cdot 6400\ \text{m}$ (for this last one, explain the complication in one sentence).

b. Write $149{,}600{,}000\ \text{km}$ (the Earth–Sun distance) in scientific notation.

c. Write $0.00000075\ \text{kg}$ (a dust grain's mass) in scientific notation.

d. Write $4.2 \times 10^{-3}\ \text{s}$ as an ordinary decimal number.

WORK

3 Decoder ring

GREEK SYMBOLS

Under each symbol, write the letter of its meaning: **(A)** an angle **(B)** "the change in" **(C)** density **(D)** wavelength **(E)** coefficient of friction **(F)** "the sum of" **(G)** torque **(H)** angular velocity

Δ _____	Σ _____	θ _____	ω _____
λ _____	μ _____	ρ _____	τ _____

Bonus: μ works a second job as an SI prefix. Name the job, and say how you can tell which job it's doing in a sentence.

Rung 2 — Base Camp: One Tool at a Time

FULL PROTOCOL · ~25 MIN

One problem, one tool: a conversion, an equation, or a single trig ratio. The protocol carries the weight.

4 The trail marshal

PROTOCOL · CONVERSION

A bike trail posts a speed limit of $30\ km/h$. A cyclist blows past a radar sign that clocks her at $9.5\ m/s$. Run the full protocol: convert the posted limit to m/s, then rule — is she over the limit, and by how much?

K	EVERY GIVEN, WITH UNITS
D	THE TARGET & ITS UNIT
E	EQUATIONS / CONVERSIONS
A	SYMBOLS FIRST · UNITS THE WHOLE WAY

5 The tunnel

D = VT · REARRANGE

A subway train cruises through a $2.7\ km$ tunnel at a steady $18\ m/s$. Run KDEA to find how long the train is inside the tunnel, in **seconds**. Two conversions and a rearrangement live inside this one — the D row tells you where you're headed.

K	EVERY GIVEN, WITH UNITS
D	THE TARGET & ITS UNIT
E	EQUATION FIRST, SYMBOLS BEFORE NUMBERS
A	SYMBOLS FIRST · UNITS THE WHOLE WAY

6 The waterslide

SOH

A waterslide is $12\ m$ long from top to bottom, dropping at $30°$ below the ground. Run KDEA to find how high the top platform sits above the splash pool.

a. Find the height.

b. **Calculator check:** what must $\sin\ 30°$ equal on a calculator in the right mode — and roughly what would a calculator left in the wrong mode whisper instead? (Run it and see.)

K	EVERY GIVEN, WITH UNITS
D	THE TARGET & ITS UNIT
E	WHICH RATIO? SOH · CAH · TOA
A	SYMBOLS FIRST · UNITS THE WHOLE WAY

Rung 3 — The Climb: Two Tools or Two Steps

CHOOSE THE TOOL, THEN CHAIN IT · ~30 MIN

Nobody hands you the ratio anymore. Naming the tool — before touching the calculator — is the skill being graded.

7 The ladder

PYTHAGOREAN

A `4.5 m` ladder leans against a wall with its foot `1.8 m` out from the base.

a. Run KDEA to find how high up the wall the ladder reaches. No angle is given — none is needed. Pick your tool accordingly.

b. Sanity-check: what relationship must your answer have with `4.5 m`, and why?

K	EVERY GIVEN, WITH UNITS
D	THE TARGET & ITS UNIT
E	WHICH TOOL — AND WHY THAT ONE?
A	SYMBOLS FIRST · UNITS THE WHOLE WAY

8 The drone delivery

FULL VECTOR

A delivery drone flies `60 m` east down a street, then turns and flies `80 m` north to the porch. Find its displacement from the takeoff point — that means **both halves of the vector**:

a. **Magnitude:** the straight-line distance to the porch.

b. **Direction:** the angle of that line, north of east.

Bonus: this triangle belongs to a famous family from class. Name the family, then write both non-right angles without touching your calculator.

K	EVERY GIVEN, WITH UNITS
D	TWO TARGETS, TWO UNITS
E	ONE TOOL PER TARGET
A	SYMBOLS FIRST · UNITS THE WHOLE WAY

9 The cable car

COMPONENTS

A cable car travels `1.4 km` along its cable, which points `18°` above the horizontal.

a. How far horizontally does it travel? (`x = r cos θ`)

b. How much altitude does it gain? (`y = r sin θ`)

c. **Check yourself:** put your two components back through `a² + b² = c²`. What should come out, and does it (within rounding)?

K	EVERY GIVEN, WITH UNITS
D	THE TARGETS & THEIR UNITS
E	EQUATIONS
A	SYMBOLS FIRST · UNITS THE WHOLE WAY

10 Grade this work

CRITIQUE

The problem: *"A van is clocked at 90 km/h in a zone posted at 25 m/s. Is it speeding?"* A student turns in, in full:

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90 × 3.6 = 324 m/s — the van is super speeding.
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a. Find at least **three** distinct violations. Be specific — name the step each one skipped or broke (protocol, units traveling, sanity check…).

b. Now do it right: run the full protocol and deliver the verdict.

K	EVERY GIVEN, WITH UNITS
D	THE TARGET & ITS UNIT
E	EQUATIONS / CONVERSIONS
A	SYMBOLS FIRST · UNITS THE WHOLE WAY

Rung 4 — Boss Level: Everything at Once

HOMEWORK · FINISH BEFORE MONDAY

Monday's last section looks like this rung. If you can climb these two with your card beside you, you're ready.

11 The zip line, all the way

INTEGRATION

A zip line cable stretches 95 m from the launch platform down to the landing anchor, angled 24° below the horizontal.

a. Find the vertical drop from launch to landing.

b. Find the horizontal span the cable covers.

c. Write your drop from (a) in **kilometers**, in scientific notation.

d. Park rules say a zip line's drop may not exceed 50 m. Does this one pass?

K	EVERY GIVEN, WITH UNITS
D	THREE TARGETS, THREE UNITS
E	ONE TOOL PER TARGET + ONE CONVERSION
A	SYMBOLS FIRST · UNITS THE WHOLE WAY · HONEST PRECISION

12 Fermi: the class-time census

ESTIMATION

Estimate: how many seconds of class time does a Cushman student sit through in one whole school year? Do not look anything up — that's the point. This problem fuses both halves of Unit 0: Fermi's method *and* the conversions that carry units through the math.

a. Break the question into three or four quantities you can reasonably guess, and write your guess for each — with units.

b. Multiply through (units travel!) and report an **order of magnitude**.

c. Which guess are you least confident in? If it were off by a factor of 2, would your order of magnitude survive?

WORK

Graded on the homework 4–3–2–1 scale — completeness and honest reasoning, not just right answers. There is no single correct number for Problem 12; there are only defensible ones. Bring Monday: this finished guide, your calculator (DEG mode!), and your handwritten 4×6 card.